

CareerStack: An Integrated Framework for Skill Gap Analysis and Economic Growth Tracking

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Abstract—The mismatch between academic output and industry requirements remains a significant barrier to global economic growth. This paper presents CareerStack, a full-stack platform designed to align with UN Sustainable Development Goal 8 (SDG-8). CareerStack provides a quantified skill-match engine, an intelligent course recommender based on gap coverage, and a financial growth tracker. Built using a Node.js/Express.js RESTful API and a MongoDB data layer, the system demonstrates an efficient, zero-framework architectural approach to career readiness. Experimental results show that the platform successfully identifies missing competencies and suggests optimized learning paths to improve employability.

Keywords—Career Development, Skill Matching, Job Recommender System, Income Tracker, RESTful API, Node.js, Express.js, MongoDB, JWT Authentication, SDG-8.

I. INTRODUCTION

The global employment landscape is undergoing rapid transformation driven by technological advancements and digital disruption. A growing mismatch exists between academic skill development and industry demands. In India, millions of engineering graduates enter the workforce annually, yet many remain underemployed due to inadequate alignment between technical competencies and skill requirements of employers. This skills gap undermines the United Nations Sustainable Development Goal 8 (SDG-8), which advocates for sustained economic growth and full employment. CareerStack responds to this challenge as a full-stack web application helping students identify skill gaps and discover learning resources.

II. LITERATURE REVIEW

Prominent platforms like LinkedIn and Coursera address career development but often lack integrated, personalized skill gap analysis combined with income tracking. Research establishing task-based models demonstrates how technological change creates persistent mismatches. Recommender systems for learning resources have explored filtering approaches; however, lightweight systems suited for resource-constrained student environments remain scarce. CareerStack fills this gap by embedding an income tracker within a career development workflow, correlating skill acquisition milestones with financial growth.

III. SYSTEM ARCHITECTURE

CareerStack follows a decoupled client-server architecture. The Presentation Tier consists of a Multi-Page Application (MPA) using vanilla JavaScript and a custom CSS design system. The Application Tier is a Node.js/Express.js REST API with modular routes for authentication, jobs, courses, and income. The Data Tier utilizes MongoDB via the Mongoose ODM for schema-based storage. Authentication is managed through JSON Web Tokens

(JWT), ensuring stateless and secure communication between the frontend and backend.

IV. METHODOLOGY

A. Skill Match Engine

The Skill Match Engine normalizes user-inputted skills and job-required skills to lowercase keywords. It uses set intersection to identify matched skills and set difference for missing skills. The match percentage is computed based on the ratio of matched to required skills. Results are categorized into levels: Strong Match ($\geq 80\%$), Moderate Match ($\geq 60\%$), Needs Improvement ($\geq 40\%$), and Low Match.

B. Course Recommendation Logic

The recommendation system ranks courses by "Gap Coverage." For each course in the database, the system counts how many of its skill tags overlap with the user's identified missing skills. Courses are then sorted descending by this coverage count, ensuring users receive the most efficient learning resources for their specific gaps.

V. EXPERIMENTAL RESULTS

The system was tested using 6 job roles and 16 curated courses. API performance metrics showed average response times under 50ms for job and course retrieval. Security tests confirmed that JWT-based route protection effectively restricted unauthorized access. The income tracker correctly visualized growth over a simulated 6-month period using Chart.js, demonstrating its effectiveness as a career progress visibility tool.

VI. CONCLUSION

CareerStack demonstrates that a lightweight full-stack application can effectively deliver career intelligence tools. By bridging information asymmetry between students and employers, the platform supports SDG-8 objectives. Future

directions include integrating Natural Language Processing (NLP) for semantic matching and automated course ingestion from external APIs.

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